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Energy-Aware Digital Twin of LEO Satellite Constellations with GNN-Based Collision Prediction

An Interdisciplinary Project Report (IDP-2025)

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In partial fulfillment of the requirements for the degree of
Bachelor of Engineering in respective departments

2025-26



RV College of Engineering®, Bengaluru
(Autonomous institution affiliated to VTU, Belagavi)



CERTIFICATE

Certified that the interdisciplinary project (IDP-2025) work titled ***Energy-Aware Digital Twin of LEO Satellite Constellations with GNN-Based Collision Prediction*** is carried out by **Khushi Choudhary (1RV22CS000)**, **P Narashimaraja (1RV22EC005)**, **Nithin M (1RV22EC006)**, **Aditya (1RV22EE007)**, **Bala N (1RV22EC007)** and **Advaith Singh (1RV22CS012)** who are bonafide students of RV College of Engineering, Bengaluru, in partial fulfillment of the requirements for the degree of **Bachelor of Engineering** in respective departments during the year 2025-26. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the interdisciplinary project report deposited in the departmental library. The interdisciplinary project report has been approved as it satisfies the academic requirements in respect of interdisciplinary project work prescribed by the institution for the said degree.

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DECLARATION

We, **Khushi Choudhary, P Narashimaraja, Nithin M, Aditya, Bala N and Advait Singh** students of sixth semester B.E., RV College of Engineering, Bengaluru, hereby declare that the interdisciplinary project titled '**Energy-Aware Digital Twin of LEO Satellite Constellations with GNN-Based Collision Prediction**' has been carried out by us and submitted in partial fulfilment for the award of degree of **Bachelor of Engineering** in respective departments during the year 2025-26.

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any degree or diploma to any other university.

We also declare that any Intellectual Property Rights generated out of this project carried out at RVCE will be the property of RV College of Engineering, Bengaluru and We will be one of the authors of the same.

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ABSTRACT

Highlights of significant contributions: One page with 3 to 4 paragraphs

Paragraph 1: Importance and relevance of Topic, reported issues and limitations of the topic in performance or computation etc, issues involved in those limitations, need for addressing those issues and a short note on how that is addressed in this report.

Paragraph 2: Objectives of this work, short note on algebraic methods used and formulations achieved, computational procedures developed. Integrated Circuit (IC).

Paragraph 3: Description of simulation procedure including SW tools used and choice of test cases. Short note on results achieved and significant highlights of improvements if any in terms of percentage , for example the architecture presented in this report shows 22 % improve in power consumption as compared to the previously reported articles. IC

Paragraph 4: The last para needed only if emulation (after simulation) or Hardware developed to validate simulation results in para 3.

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LIST OF TABLES

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Introduction

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INTRODUCTION

0.1 Motivation

Low-Earth-orbit constellations now host thousands of satellites, increasing conjunction risk. This project aims to provide a proactive digital-twin solution.

0.2 Objectives

- High-fidelity N-body physics with mass-momentum conservation.
- Real-time TLE ingestion.
- GNN-based collision-risk prediction.
- Interactive UI for visual risk assessment.

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System Architecture

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SYSTEM ARCHITECTURE

0.3 Simulation Core

The simulator is built in Rust using the `glam` crate for vector math and `wgpu` for GPU rendering. The core loop resides in `src/simulation/state.rs` and advances the system using a Velocity-Verlet integrator.

0.4 Data Ingestion

Two-Line Element (TLE) files are parsed by the `tle_parser` module and loaded into `SimState.UpdatedT`

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Physics Engine Enhancements

PHYSICS ENGINE ENHANCEMENTS

0.5 Mass-Momentum Conserved Fragmentation

During a collision the debris_{gen} module creates a cloud of fragments. The total mass of the frag

0.6 Numerical Integration

The simulation advances using a Velocity-Verlet scheme with adaptive time steps to maintain stability during high-speed events.

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GNN-Based Collision Prediction

GNNBASED COLLISION PREDICTION

0.7 Graph Construction

A proximity graph is built each tick with an 800 km edge threshold. Nodes represent satellites; edges encode relative kinematics.

0.8 Feature Engineering

Features per edge include distance, relative speed, convergence rate, mass ratio, and altitude difference.

0.9 Message Passing Heuristic

A single message-passing iteration aggregates neighbour features and produces a risk score in the range $[0,1]$ for each satellite pair.

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User Interface Visualization

USER INTERFACE VISUALIZATION

0.10 HUD Overview

The UI is built with egui. It provides three panels: a risk table (top-right), a selected-object info panel (bottom-right), and a colour legend (bottom-left).

0.11 Red Tracker Arrow

A foreground-layer arrow points to the predicted impact location and bounces slightly to attract attention.

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Results & Evaluation

RESULTS & EVALUATION

0.12 Scenario Description

We simulate a collision between ISS-Zarya, Iridium-33 and Cosmos-2251. The engine generates fragments while maintaining physical conservation laws.

0.13 Prediction Accuracy

The GNN predictor identified high-risk pairs with high confidence seconds before physical contact.

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Discussion & Future Work

DISCUSSION & FUTURE WORK

0.14 Discussion

The integration of GNNs allows for proactive space traffic management.

0.15 Future Work

Future enhancements include loading trained models via ONNX and scaling to thousands of objects.

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Conclusion

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CONCLUSION

We have developed a high-fidelity digital twin of LEO satellite constellations. The system demonstrates the feasibility of real-time collision prediction using GNNs.

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